Project 2

Congratulations on completing your second unit of training with MCB consulting group. You now get the opportunity to complete a consulting project for our client Shop.com.mm, a shopping conglomerate headquartered in Myanmar. You are tasked with helping them address one of the following pain points. Remember – clients are good at identifying problems but are not necessarily able to identify the best way to solve that problem. Deliver the analysis they have requested but make sure to provide relevant recommendations based on that analysis.

1. **Assessing Loyalty Program Effectiveness**: Shop.com.mm has invested heavily in its customer loyalty program. Currently, members receive additional benefits such as early access to new products and members-only sale pricing. However, upper management has requested an in-depth analysis about whether the loyalty program is benefitting the platform. Your job is to answer this question (*e.g.*, through a key drivers analysis of an outcome of interest such as the total revenue per order) and to provide recommendations on whether to maintain or revise the current loyalty program.
2. **Quantifying Customer Satisfaction**: Shop.com.mm collects customer ratings after order fulfillment. There are many factors that could affect this rating, such as the shopping/ordering experience, fulfillment efficacy, and product quality and satisfaction. Management has noticed that customer ratings vary widely even across orders that have similar sizes and product types. Your job is to assess which factors are driving customer ratings and to provide recommendations accordingly. In other words, what levers can Shop.com.mm use to increase customer ratings?

Shop.com Data Dictionary

Invoice ID: invoice ID number. Each corresponds to a unique order.

Branch: branch location at which the order was fulfilled.

City: city in which the branch is located.

Customer type: whether the customer is a member of the loyalty program or just a standard customer.

Gender: gender of the customer.

Product line: corresponding product line for the ordered product.

Unit price: price per unit ordered.

Quantity: number of units ordered.

Total: total purchase price including tax.

Date: date on which the order was placed.

Time: time of day the order was placed.

Payment: payment type.

Rating: customer satisfaction rating (collected after the product was received).

Expectations

1. Review the diagram of the ***Marketing Analytics Process***. Note that we stopped in the VISUALIZATION stage in our last project. For this project, it is expected that you will go through the full process from IMPORTING to TRANSFORMING, VISUALIZING, and MODELING the data, before COMMUNICATING your results. This means the depth of your work for this project (Project 2) should be *more* than what you did for Project 1 (*i.e.*, you are building on the amount of work you did before by adding model).

Deliverables

1. Technical Report
    a. A rendered word document containing all code used for importing, tidying, visualizing, summarizing, and interpreting data. Code should contain comments and clearly show the work that was done, *e.g.,* plots should be fully rendered with labeled axes and titles. Document your full decision process so that our Senior Analytics Partners can follow along with your work and replicate your results. Your code must be able to run on their computers, so remember to test all your code in a fresh R session before submission.
2. Executive Summary
    a. A one- to two-page document summarizing the work that you completed. The summary should do the following:
        i. Identify the problem.
        ii. Briefly describe the data used in real-world terms (no coding jargon).
        iii. Summarize the analysis you conducted and your results.
        iv. Identify the relevance of your results to Shop.com.mm's business practices.
        v. Make recommendations on how to use the results effectively and describe any further analyses you think would be worthwhile.